CP–7831

Sub. Code
11H

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Science

HINDI PAPER – I – STORY, NOVEL, GRAMMAR AND TRANSLATION

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

सभी प्रश्नों के उत्तर संक्षेप में लिखिए।

1. प्रेमचन्द के किन्हीं चार उपन्यासों का नाम लिखिए।

2. पंडित परमसुख चौबे का संक्षिप्त परिचय दीजिए।

3. हज़ारासिंह का संक्षिप्त परिचय दीजिए।

4. अंतिम मुलाकात में प्रकाश ने अपने बेटा गोपाल से क्या कहा?

5. वकील उदयभानुलाल को किसने मार डाला और क्यों?

6. निर्मला की शादी किससे हुई? और उसकी ननद का नाम क्या है?

7. मुन्शी जी निर्मला के पास अपनी वीरता की डींग किस प्रकार मारने लगे?

8. सोहन की मृत्यु कैसे हुई?

9. लिंग बदलिए:-

(a) देव (b) बूढा

(c) ऊँट (d) धोबी

10. वचन बदलिए।

(a) घडी (b) कन्या

(c) सडक (d) वस्तु

Part B $(5 \times 5 = 25)$

सभी प्रश्नों के उत्तर दीजिए।

11. (a) आनन्दी का संक्षिप्त परिचय दीजिए।

या

(b) 'उसने कहा था' इस कहानी में किसने, किससे क्या कहा था?

12. (a) रामू की माँ का संक्षिप्त परिचय दीजिए।

या

(b) प्रकाश ने क्या पाप किया? क्यों जेल गये?

13. (a) एकवचन से बहुवचन बनाने के नियम बताइए।

या

(b) पुल्लिंग से स्त्रीलिंग बनाने के नियम बताइए।

14. (a) मोटेराम का चरित्र-चित्रण कीजिए।

या

(b) सुधा का चरित्र-चित्रण कीजिए।

15. (a) मंसाराम का चरित्र–चित्रण कीजिए।

या

(b) निर्मला में लेखक ने हिन्दू समाज की किन किन कुरीतियों का चित्रण किया है?

Part C $(3 \times 10 = 30)$

सभी प्रश्नों के उत्तर दीजिए।

उत्तर विस्तार से लिखिए।

16. (a) 'प्रायश्चित' कहानी का सारांश लिखिए।

या

(b) निर्मला उपन्यास की समस्याओं पर समीक्षा कीजिए।

17. (a) संज्ञा किसे कहते हैं? उनके भेदों को उदाहरण सहित समझाइए।

या

(b) सर्वनाम किसे कहते हैं? उनके भेदों को उदाहरन सहित समझाइए।

18. अंग्रेजी में अनुवाद कीजिए।

(a) गाँधीजी ने एक बार कहा था, 'धन गया तो कुछ नहीं गया, स्वस्थ्य गया तो कुछ गया, मगर चरित्र गया तो सब कुछ गया'। इसका मतलब है, चरित्र ही मनुष्य की असली संपत्ति है।

वही समाज श्रेष्ठ है और वही राष्ट्र सुखी है जिसकी जनता का चरित्र अच्छा हो। आज के बालक कल के नागरिक हैं। इसलिए उनके चरित्र का निर्माण करना समाज और राष्ट्र का पहला कर्तव्य है।

या

(b) कौए हिन्दुस्तान के प्राय: सभी हिस्सों में पाये जाते हैं – मैदानों में भी और पहाड़ी क्षेत्रों में भी। सिक्किम में 18, 14 हज़ार फुट की ऊँचाई पर भी ये पाये गये हैं, पर पहाड़ी कौए की रूपरेखा, कद आदि में मैदानी कौओं से काफ़ी फ़र्क आ जाता है। जब–तब एक ख़ास प्रकार का कौआ भी कही–कही देखा गया है, जिसका सारा बदन सफ़ेद होता है, पर यह कम ही देखने को मिलता है।

CP–7832

Sub. Code
11F

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Science

Part I — FRENCH

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Complétez avec les formes d'avoir

   (a) Moi, j' ———— sept jeux vidéo.

   (b) Et mon papa, il ———— un avion.

2. Remplacez ici par à + ville

   (a) Vous arrivez ici quand? (Reims)

   (b) Tu es ici pour trois jours. (Toulouse)

3. Traduisez en français

   (a) Good night

   (b) Happy journey

4. Complétez les phrases

   (a) a.je ———— (japonais/Nagoya)

   (b) Moi, je ———— (espagnol/Séville)

5. Faites l'addition (toutes en lettres)

   (a) 10 + 11 =

   (b) 3 + 2 =

6. Mettez à l'impératif à la 2nd personne du pluriel

   (a) Ecouter

   (b) Parler

7. Complétez par du, de la, des

   (a) ———— pommes

   (b) ———— salade

8. Donnez l'heure (en toutes letters)

   (a) 1hoo

   (b) 04h o5

9. Complétez par quand ou combien

   (a) Le plan coute ———— ?

   (b) c'est ———— la fête Laurent?

10. Utilisez Mon, Ma, Meg

   (a) ———— parents.

   (b) ———— oncle.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Ecrivez le jour de la semaine.

Ou

(b) Mettez les verbes au présent :

(i) Vous (arriver) demain?

(ii) Tu (être) étudiant?

(iii) Je (rester)10 jours.

(iv) Tu (habiter) a Montréal?

(v) Ils (vérifier) les réponses.

12. (a) Complétez par l'impératif a la 2nd personne du singulier et du pluriel :

(i) (Ecouter) encore l'enregistrement.

(ii) (cocher) les bonnes cases.

Ou

(b) Compétez par vouloir et pouvoir :

(i) Je _____ entrer? (pouvoir)

(ii) qu'est –ce vous_____ au fait ? (vouloir)

(iii) Nous ———— être à neuf heures (pouvoir)

(iv) Nous ———— voir le directeur, s'il vous plait (vouloir)

(v) Je ———— faire?

13. (a) Complétez par au, à, la, aux, du, de, la, des :

(i) Je vais acheter deux baguettes ———— boulangerie. (f).

(ii) l'école est à gauche ———— rue Jules Verne. (f).

(iii) Le magasin bio se trouve à cote ———— lycée. (m).

(iv) Tu fais le menage ———— appartement ? (m).

(v) Tu aimes la tartes ———— citron ? (m.).

Ou

(b) Complétez les mots avec :- an/am, en/em, ou, on/om

(i) professi ————

(ii) Rep ———— se

(iii) Naiss ———— ce

(iv) Dim ———— the

(v) N ————

14. (a) Faites des phrases :

(i) sport

(ii) vin

(iii) rose

(iv) photo

(v) bonsoir

Ou

(b) Faites des phrases :

(i) Parler

(ii) Chocolatés

(iii) Ville

(iv) France

(v) Long

15. (a) Complétez avec Est-ce que(ou) qu'est-ce que :

(i) ———— tu achetés au supermarché?

(ii) ———— vous aimez voyager?

(iii) ———— tu restes ici ?

(iv) ———— vous préférez le chocolat blanc ou le chocolat noir?

(v) ———— tu fais?

Ou

(b) Complétez par quand ou combien, selon cas :

(i) Tu vas ———— au concert?

(ii) ———— c'est, les trois timbres?

(iii) Le plan coute ————?

(iv) Tu es à Nice ————?

(v) C'est ———— la fête de Laurent?

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Ecrivez les nombres de 1 à 25.

Ou

(b) Ecrivez les nombres de 25 à 50.

17. (a) Décrivez votre ville?

Ou

(b) Ecrivez dix couleurs en française?

18. (a) Décrivez un personnage célèbre?

Ou

(b) Décrivez votre famille?

CP–7833

Sub. Code
12

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Science

Part II — PROSE AND COMMUNICATION SKILLS

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Who is compared to the man who stumbles through his lessons?

2. On whom our hopes are pinned according to Indira Gandhi?

3. What did D'Artagnan carry with himself?

4. What punishment did the teacher give the boy?

5. What did Captain Marvel and Walter Scott provide?

6. How does Sarojini Naidu humbly describe herself?

Fill in the blanks with appropriate form of verbs.

7. I ———————— here since 1999.(live)

8. He ———————— tea daily. (drink)

Change into passive voice.

9. She is reading the novel.

Use appropriate modal verb.

10. ———————— God bless you!

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) What is the real objective of Education according to Livingstone?

Or

(b) Why does Indira Gandhi say that youth are the largest repository of power?

12. (a) Why was Balfour agonized when made a speech?

Or

(b) Bring out the humour in R.K. Narayan's Crime and Punishment.

13. (a) Why did Margaret Atwood prefer animal stories?

Or

(b) What are the three visions that Sarojini Naidu talks of?

14. (a) Write on usage of Present continuous tense.

Or

(b) Write on usage of past continuous.

15. (a) Use appropriate Modal Verbs:

(i) We ———————— Vote only after we complete eighteen.

(ii) A good teacher ———————— make even boring lessons interesting.

(iii) The sky is overcast. It ———————— rain.

(iv) He said that it ———————— rain that day.

(v) ———————— I close the doors?

Or

(b) Use appropriate articles:

I was reading ———————— novel yesterday. It was written by ———————— Telugu novelist ———————— hero saved ———————— child from the clutches of ———————— tiger.

Part C (3 × 10 = 30)

Answer **all** questions, choosing either (a) or (b).

16. (a) Write an essay on how to attain Excellence in education emphasized by Livingstone.

Or

(b) Bring out the humour in the allusion to Stevenson and his Modestine.

17. (a) What are the three kinds of courage put forth by Indira Gandhi?

Or

(b) Write an essay on the books that Margaret Atwood enjoyed reading when she was a little girl. What reasons does she give for her preferences?

18. (a) Fill in the blanks with appropriate verbs:

(i) My sister ———————— (clean) her room just now.

(ii) I don't know why he ———————— (kill) her yesterday.

(iii) We ———————— (sing) here at six o'clock this evening.

(iv) She ———————— (work) here since her childhood.

(v) Mani ———————— (read) this book for the past three hours.

(vi) They ———————— (wait) for her since 8 o'clock.

(vii) By this time next month, I ———————— (build) this house.

(viii) At this time tomorrow, she ———————— (play) here.

(ix) We ———————— (discuss) the matter already in detail.

(x) He ———————— (write) three essays so far.

Or

(b) Write an essay on articles. Give examples.

———————

CP–7834

Sub. Code
13

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aero Nautical Science

MATHEMATICS – I

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Find the rank of the matrix $A = \begin{pmatrix} -1 & 2 & 3 \\ 2 & -4 & 6 \\ 1 & -1 & -2 \end{pmatrix}$.

2. Prove that the eigen value of the matrix A^{-1} are $\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \ldots, \frac{1}{\lambda_n}$ if the eigen values of A are $\lambda_1, \lambda_2, \ldots, \lambda_n$.

3. If a straight line makes angles 60° and 45° with X and Y axes respectively, find the angle made by the line with Z-axis.

4. Find the angle between the planes $2x + 3y + 4z - 2 = 0$ and $x - 2y + z + 4 = 0$.

5. Find the radius of the curvature of $y = 2x^2$ at (1, 1).

6. Find centre of the curvature $y = x^3$ at (1, 1).

7. Verify Euler's Theorem, when $u = x^3 + y^3 + z^2 + 2xyz$

8. Find the critical points of $y = (x-1)^2(x+2)$.

9. Find I.F of the differential equation $\frac{dy}{dx} + y\cos x = \frac{1}{2}\sin 2x$.

10. Find the P.I. of $(D^2 + 5D + 6)y = e^x$.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Find the rank of the matrix $A = \begin{pmatrix} 1 & 2 & 1 & 2 \\ 1 & 3 & 2 & 2 \\ 2 & 4 & 3 & 4 \\ 3 & 7 & 4 & 6 \end{pmatrix}$.

Or

(b) Verify Cayley Hamilton Theorem for the matrix $A = \begin{pmatrix} 1 & 0 & -2 \\ 2 & 2 & 4 \\ 0 & 0 & 2 \end{pmatrix}$ and find A^{-1}.

12. (a) Find the image of the point (1,3,4) in the plane $2x - y + z + 3 = 0$.

Or

(b) Find the equation of the sphere passing through the circle $x^2 + y^2 + z^3 - 2x + 4y + z - 2 = 0,\ x + 3x - z = 4$.

13. (a) Prove that the radius of curvature of the curve $xy^2 - a^2 - x^2$ at $(a, 0)$ is $\frac{3a}{2}$.

Or

(b) Using the formula $\rho = r\frac{dr}{d\theta}$, find the radius curvature at any point on the curve $r^2 = a^2 \sin 2\theta$.

14. (a) Expand $e^x \cos y$ at (0,0) up to the third degree.

Or

(b) Use Taylor's formula to expand the function f defined by $f(x, y) = x^3 + y^3 + xy^2$ in powers of $(x-1)$ and $(y-2)$.

15. (a) Solve $\frac{dy}{dx} - \frac{xy}{1-x^2} = \frac{1}{1-x^2}$.

Or

(b) Solve $(D^2 - 5D + 6)y = 2e^{2x}$.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Find the eigen values and the eigen vectors of the matrix $A = \begin{pmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{pmatrix}$.

Or

(b) Reduce the Quadratic form $Q = x_1^2 + 3x_2^2 + 3x_3^2 - 2x_2x_3$ to a canonical form by a suitable transformation of the variable.

17. (a) Find the shortest distance between the lines $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and $\frac{x-2}{3}=\frac{y-4}{4}=\frac{z-5}{5}$ and its equations.

Or

(b) Show that evolute of the parabola $y^2 = 4ax$ is the curve $27ay^3 = -4(x-2a)^3$.

18. (a) Maximize $z = xy$ subject to the condition $(x-2)\,(y-4) = 18$.

Or

(b) Solve, by the method of variation of parameters, the equation $\frac{d^2y}{dx^2} + a^2 y = \sec ax$.

CP–7835

Sub. Code
14

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Science

WORKSHOP PRACTICES

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define Fire. Mention various sources of Fire.

2. Discuss the arrangement of tools used in Air Frame Bay.

3. Discuss any five common hard tools used in Aircrafts construction.

4. Why is Calibration of equipment necessary in maintenance Bay?

5. Define Least Count of micrometer.

6. Mention various types of fits used.

7. Define the Principle of Gas Welding.

8. Define Lathe.

9. Discuss the location of workshop of maintenance Bay.

10. Draw a neat sketch of micrometer and label the parts.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Discuss the Safety precautions to be followed while handling Electricity in Aircraft construction.

Or

(b) Safety Precaution in Machine shop.

12. (a) Explain various types of fire extinguishers with symbols.

Or

(b) Explain the working of column type milling machine with a sketch.

13. (a) Explain various types of Hammers with sketch and Define Lathe.

Or

(b) Draw a neat sketch of Internal and External Micrometer and Explain.

14. (a) Write short notes on Standards of Workmanship in Aircraft Construction.

Or

(b) Explain various tools used in Gas Welding Plant.

15. (a) Explain various types of gears with sketch.

Or

(b) Explain the Standard method for Checking Shafts, Bearings.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Explain various safety precautions to be followed while handling oxygen system.

Or

(b) Safety precautions to be followed while using oils and chemicals.

17. (a) Explain various types screw drivers with sketches.

Or

(b) Explain various parts of Lathe with diagrams.

18. (a) Explain the applications of Lathe, Milling machines.

Or

(b) Explain Limits, Fits and Tolerance.

———————

CP–7836

Sub. Code
15

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Sciences

BASIC ELECTRICITY AND ELECTRONICS

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define Norton's Theorem.

2. Define instrument transformers.

3. Illustrate the difference between DC shunt and series motor.

4. Write the EMF equation of DC generator.

5. Define universal motor.

6. Draw the diagram for single phase induction motor.

7. Define UJT.

8. Define Tolerance in transistor.

9. Write short notes for Shockley diode.

10. Define SMPS.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Describe the effect of series and parallel resistances.

Or

(b) Describe the spiral transformer with neat diagram.

12. (a) Describe the EMF equation of DC generator and their constructions.

Or

(b) Describe DC shunt motor starter.

13. (a) Explain the production of rotating magnetic field in 3 phase induction motor.

Or

(b) Explain the production of rotating magnetic field in single phase induction motor.

14. (a) Write the short notes for MOSFET and UJT.

Or

(b) Define transistor operation and their configurations.

15. (a) Write the characteristics of SCR.

Or

(b) Describe the difference between SMPS and UPS.

Part C $(3 \times 10 = 30)$

Answer **all** questions.

16. (a) Explain briefly Core/ Shell transformer with neat diagram.

Or

(b) Explain briefly Kirchhoff's law and Thevinin theorem.

17. (a) Give brief notes for FET and JFET and their types.

Or

(b) Give brief notes for single phase induction motor.

18. (a) Write the characteristics of TRIAC and DIAC and Shockley Diode.

Or

(b) Give brief notes for SMPS and UPS with neat diagram.

CP–7838

Sub. Code
21H

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Science

Part I – HINDI – PROSE, GRAMMAR AND TRANSLATION

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

सभी प्रश्नों के उत्तर संक्षेप में लिखिए।

1. वैज्ञानिक युग में भारत का दायित्व क्या है?

2. जीवन में घृणा का स्थान पाठ का लेखक कौन है? और यह पाठ किस विधा की रचना है?

3. लेखक ने रजिया को अन्तिम बार किस रूप में देखा?

4. मक्रील कहानी की भाषा पर प्रकाश डालिए।

5. भ्रमण के लिए नक्शे से क्या मदद मिलती है?

6. भारत के सम्बन्ध में गाँधी जी का कल्पना क्या थी?

7. क्रिया–विशेषण किसे कहते हैं?

8. परिमाणवाचक क्रिया–विशेषण किसे कहते हैं? उदाहरण सहित समझाइए।

9. संभाव्य भविष्यत् काल किन अर्थों में प्रयुक्त होता है?

10. नामधातु किसे कहते हैं? उदाहरण सहित समझाइए।

Part B $(5 \times 5 = 25)$

सभी प्रश्नों के उत्तर दीजिए।

संदर्भ सहित व्याख्या कीजिए।

11. (a) उस नैतिकता पूर्ण जन-शक्ति के आघात से, ब्रिटिश साम्राज्य चूर-चूर हो गया।

या

(b) 'रमता राम' इसीलिए कहते हैं कि जो रमता नहीं, वह राम नहीं। टिकना तो मौत है।

12. (a) मेरे दुख का दुर्दम वेग असह्य हो जब उछल पड़ता है, तब भी संसार उसे विनोद का ही साधन बैठता है।

या

(b) फिर जरा-सा रुककर बोली-सुना था, आप यहीं रहते हैं। कहाँ रहते हैं मालिक?

13. (a) अगर आज़ वह धूर्तता का परित्याग कर दे, तो हमारी घृणा भी जाती रहेगी।

या

(b) 'कोस-कोस पर बदले पानी, चार कोस पर बानी'।

व्याकरण :-

14. (a) 'ने' विधि का प्रयोग समझाइए।

या

(b) वर्तमान काल के भेदों को समझाइए।

15. (a) भूत काल के भेदों को समझाइए।

या

(b) भविष्यत् काल के भेदों को समझाइए।

Part C $(3 \times 10 = 30)$

सभी प्रश्नों के उत्तर दीजिए।

उत्तर विस्तार से लिखिए।

16. (a) 'बहता पानी निर्मला' पाठ का सारांश लिखिए।

या

(b) 'महात्मा गाँधी' पाठ का सारांश लिखिए।

17. (a) 'कारक' किसे कहते हैं? उनके भेदों को उदाहरण सहित समझाइए।

या

(b) 'क्रिया' किसे कहते हैं? उनके भेदों को उदाहरण सहित समझाइए।

18. हिन्दी में अनुवाद कीजिए।

(a) We are all citizens of India. Each of us is a member of one great community, which includes every Indian. We should be grateful for this privilege and the thought of it should always prevent us from doing anything unworthy. The citizen has both rights and duties. Some people think only of their rights. They forget their duties on account of their selfishness.

या

(b) Many people say that the climate of England and that of Kodaikanal are almost the same. You do not have as much heat at Kodaikanal, as you have at Ooty during the day-time. The temperature at night is also not so cold as at Ooty. The natural scenery of Kodaikanal is just like that of England. But on account of better travel facilities more people go to Ooty. Hence there is more calm at Kodaikanal than at Ooty.

CP–7839

Sub. Code
21F

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Science

Part I — FRENCH II

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Complétez par savoir

   (a) Laurence ne _____ pas compter, elle est petite !

   (b) Vous ———— aller à la gare?

2. Mettez les verbes au participe passe

   (a) Regarder

   (b) Remplir

3. Ecrivez l'infinitif des verbes

   (a) Aimé

   (b) Dit.

4. Mettez les mots au pluriel

   (a) Un jeu.

   (b) Le tableau.

5. Remettez les mots dans l'ordre

   (a) beaucoup de courses / on/ à / faire/ a

   (b) ils / un billet de train / à / retirer / ont

6. Mettez à la forme négative

   (a) Il pane à son frère.

   (b) Nous restons une semaine.

7. Complétez avec un, une, des

   (a) _____ gomme

   (b) _____ sac

8. Donnez le contraire

   (a) grand

   (b) petit.

9. Complétez la liste avec l'adjectif de couleur correspondant

   (a) Le ciel avec des nuages est ————

   (b) Le drapeau est bleu, ————

10. Complétez la liste avec l'adjectif de couleur correspondant

   (a) Le ciel avec des nuages est ————

   (b) Le drapeau est bleu, ————

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Reliez les deux phrases avec parceque.

   (i) Je suis en retard ma voiture, ma voiture est en panne.

   (ii) Il fait froid ici, les fenêtres sont ouvertes.

(iii) Lire, c'est bien, on apprend beaucoup de choses.

(iv) Il faut partir tôt demain, il y a beaucoup de route a faire.

(v) Tu fais une soirée, c'est ton anniversaire.

Ou

(b) Remettez les mots dans l'ordre et complétez par : contre

(i) Il la porte. était

(ii) L'enfant, lance, le mur, la balle

(iii) Les chaises, la table, sont

(iv) Ils tous, sont moi

(v) La voiture, un arbre, est allée

12. (a) Remettez les mots dans l'ordre

(i) Le, Beauvais à 15 h 00. part, pour, train.

(ii) Devez, pour, vous, présenter, votre demande, mardi.

(iii) On arriver, pour, doit, 19 h 00.

(iv) Pour, j' ai, raison, mon ami

(v) Sommes, pour. nous, la paix

Ou

(b) Posez des questions avec pourquoi et donnez des explications avec parceque

(i) tu es triste/Gaston ne téléphone pas

(ii) il y a beaucoup de monde devant le museé/ aujourd'hui, c'est gratis.

(iii) ils ne viennent pas/ils sont en vacances

(iv) le vent est froid / il vient du nord

(v) je reste ici/je suis bien.

13. (a) Complétez par la bonne forme verbale, d'après les indications.

(i) A quelle heure tu ———— demain? (finir, présent)

(ii) Monsieur vous avez ————? (choisir, part, Passé)

(iii) Vous ———— être plus gentils (pouvoir, conditionnel)

(iv) Tu ———— ou se trouve la rue saint-Augustin ? (savoir, présent)

(v) J' ———— du pain (manger, imparfait)

Ou

(b) Refaites avec les articles définis

(Change indefinite articles into defenite articles)

(i) unhomme -

(ii) desgarcons -

(iii) des eleves -

(iv) des livers -

(v) un oiseau -

14. (a) Faites des phrases

(i) grand

(ii) tout

(iii) téléphone

(iv) village

(v) beaucoup.

Ou

(b) Faites des phrases

(i) riche

(ii) bien

(iii) jamais

(iv) venir

(v) musée

15. (a) Mettez les verbes au passe récent.

(i) Elle arrive a paris.

(ii) Nous rencontrons le directeur.

(iii) Ils rentrent du canada.

(iv) On ecoute la météo à la radio.

(v) Vous téléphonez à paul?

Ou

(b) Complétez avec vivre

(i) Mes parents ———— en province.

(ii) Ou est allé ———— clément?

(iii) Nous ———— notre travail.

(iv) Fabienne ———— a la campagne.

(v) Vous ———— ou, monsieur?

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Décrivez un pays France?

Ou

(b) Présentez-vous?

17. (a) Décrivez des monuments de paris?

Ou

(b) Décrivez votre maison?

18. (a) Dérivez des monuments de paris?

Ou

(b) Décrivez votre maison?

CP–7840

Sub. Code
22

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Science

Part II – PROSE, EXTENSIVE READING AND COMMUNICATION SKILLS

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are the news items attracted Kalam in Tel Aviv?

2. Name some of the duties that the artist, as a citizen, has to the community.

3. How does Haldane show that our attitude to disease is highly irrational?

4. What did Gandhi expect from us? Do we measure up to his expectations?

5. What will amaze a visitor from another planet?

6. From what hints in the story do you learn that Bob has become rich?

7. How did the conjuror treat the hat and the spectacles of the quick man?

8. Why was the astrologer reluctant to read his palm?

9. What is Subordinate clause?

10. What is Superlative Degree? Give example.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) What are the three visions of Kalam?

Or

(b) How does Nehru blame himself?

12. (a) Why can the artist not fit into a totalitarian set-up?

Or

(b) What were Robert Lynd's experiences as an arguer in school and college?

13. (a) Write a character sketch of Silky Bob.

Or

(b) How does the astrologer tackle Guru Nayak?

14. (a) Narrate how the Quick Man ruined every trick of the conjuror.

Or

(b) Write the importance of Phrase in Simple sentence.

15. (a) What are the rules for changing Pronouns in Direct to Indirect Speech?

Or

(b) Role of Adjectives in Comparative degree.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) What are the faults pinpointed by Kalam on Indians? How will you rectify it?

Or

(b) Sum up Plato's contradictory views on the artist's role in society.

17. (a) How would you describe Jimmy and Bob's personalities in the story?

Or

(b) How does R.K. Narayan expose superstitious belief of villagers?

18. (a) What are the kinds of Conjunctions? Explain in detail.

Or

(b) In what circumstances the rules for changing tenses could not be applicable while transforming sentences from Direct speech to Indirect speech? Write with Examples.

———————

CP–7841	Sub. Code
	23

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Science

MATHEMATICS — II

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Evaluate : $\int_0^a \int_0^b (x^2 + y^2)\,dx\,dy$.

2. Evaluate : $\int_0^1 dx \int_0^2 dy \int_1^2 x^2 yz\,dz$.

3. Find $\nabla\left(\frac{1}{r}\right)$.

4. Show that $\vec{f} = x^2 z\vec{i} + xyz^2\vec{j} - xz^3\vec{k}$ is irrotational.

5. Test the analyticity of the function $f(z) = e^x(\cos y + \sin y)$.

6. Evaluate : $\int \frac{dz}{z-a} = 2\pi i$.

7. Find $L\left(te^{-2t}\right)$.

8. Find $L^{-1}\left[\frac{s-3}{(s-3)^2+4}\right]$.

9. If mean is 26.5 and mode is 28, find the median using exponential relation.

10. If the regression co-efficient of x on y is $\frac{9}{20}$ and regression co-efficient of y on x is $\frac{4}{5}$. Then Find the Correlation co-efficient.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Evaluate $\int_0^{\frac{\pi}{2}} \int_0^{a\cos\theta} r^4 dr\, d\theta$.

Or

(b) Change the order of integration and evaluate $\int_0^1 \int_0^{2\sqrt{x}} x^2\, dy\, dx$.

12. (a) Find the value of 'a' if $\vec{f} = (axy - z^2)\vec{i} + (x^2 + 2yz)\vec{j} + (y^2 - axz)\vec{k}$ is irrotational.

Or

(b) If $\vec{f} = x^2\,\vec{i} + xy\,\vec{j}$ find $\int_C \vec{f} \cdot \overrightarrow{dr}$ where C is curve joining (0, 0) to (1, 1) along the parabola $y^2 = x$.

13. (a) Find the bilinear transformation that maps the points $z_1 = -i, z_2 = 0, z_3 = i$, into the points $\omega_1 = -1, \omega_2 = i, \omega_3 = 1$ respectively.

Or

(b) Find the Analytic function $f(z) = u + iv$ if $u + v = \dfrac{x}{(x^2 + y^2)}$ and $f(1) = 1$.

14. (a) Find $L^{-1}\left[\dfrac{1}{s(s+1)(s+2)}\right]$.

Or

(b) Find $L^{-1}\left[\dfrac{1}{s^2 + 4^2}\right]$.

15. (a) Compute the mean median and mode for the following data :

Marks :	10-15	15-20	20-25	25-30	30-35	35-40	40-45	45-50
No.of Students :	2	28	125	270	303	197	65	10

Or

(b) Calculate Quartile Deviation for the following data :

Marks :	0-5	5-10	10-15	15-20	20-25	25-30
No.of Students :	4	6	8	12	7	4

Part C $(3 \times 10 = 30)$

Answer **all** questions.

16. (a) Change the order of integration, $\int_0^a \int_0^a (x^2 + y^2)\,dy\,dx$ and hence evaluate the same.

Or

(b) Evaluate $\iiint xyz\,dx\,dy\,dz$ taken through the positive octant of the sphere $x^2 + y^2 + z^2 = a^2$.

17. (a) Evaluate $\iint \vec{F} \cdot \vec{n}\, ds$ if $\vec{F} = (x + y)\vec{i} + x\vec{j} + z\vec{k}$ is the surface of the cube boundary by the planes $x = 0, x = 1, y = 0, y = 1, z = 0$ and $z = 1$.

Or

(b) Show that the function $u = 2xy + 3y$ is harmonic and find the corresponding analytic function. Find its conjugate.

18. (a) Solve $\frac{d^2y}{dt^2} + 4\frac{dy}{dt} - 5y = 5$ given that $y = 0, \frac{dy}{dx} = 2$, when $t = 0$ using laplace transformation.

Or

(b) Calculate the coefficient of correlation for the following height of fathers (X) and heights of sons (Y).

X:	65	66	67	68	69	70	71	72
Y:	67	68	65	68	72	72	69	71

CP–7842

Sub. Code
24

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Science

ENGINEERING MECHANICS AND STRENGTH OF MATERIALS

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define Moment of a couple.

2. State parallelogram law of Forces.

3. Define velocity ratio and Mechanical Advantages.

4. Define simple Harmonic Motion in Dynamics.

5. Define centre of Gravity.

6. Define perfect frame.

7. Define stress and strain.

8. Define Poisson's Ratio.

9. Define Impulse momentum equation.

10. Define Co-Planar and Concurrent Forces in Mechanics.

Part B (5 × 5 = 25)

Answer **all** questions.

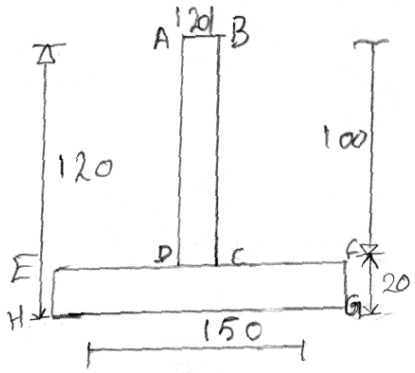
11. (a) Explain fundamental units and derived units.

Or

(b) Explain resolution of forces on a system.

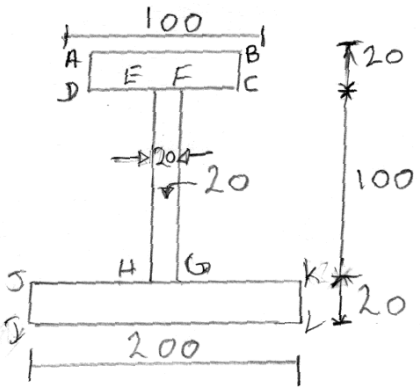
12. Determine the centre of gravity of the figure shown below :

(a)



Or

(b)



13. (a) Write short notes on Laws of Friction.

Or

(b) Explain the principle of Belt Friction.

14. (a) Explain various types of Frames.

Or

(b) Explain resolution of forces by method of joints.

15. (a) Explain various types of beams and loads.

Or

(b) Explain Shear Force and Bending moment diagram.

Part C $(3 \times 10 = 30)$

Answer **all** questions.

16. (a) Explain Composition of Forces and Couple lever arm principle.

Or

(b) Explain the principle of virtual work.

17. (a) Derive the procedure to derive centre of Gravity of the body.

Or

(b) Moment of Inertia of the body.

18. (a) Derive the equation of efficiency for Screw Jack Friction.

Or

(b) Explain Elastic Constants.

CP–7843

Sub. Code
25

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Science

ENVIRONMENTAL STUDIES

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Biosphere

2. Environmental awareness

3. Deforestation

4. Overgrazing

5. Trophic level

6. Hotspots

7. Minamata disease

8. Green house effect

9. Decibel

10. Endemic species.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Enumerate the need for public environmental awareness.

Or

(b) Discuss the scope of environmental studies.

12. (a) Write a account on forest resources.

Or

(b) Discuss the impact of modern day agriculture on environment.

13. (a) Explain the biogeographical classification of India.

Or

(b) Describe the types of ecological pyramids.

14. (a) Give an account on thermal pollution.

Or

(b) Enlist the air pollutants.

15. (a) Describe the biodiversity at various levels.

Or

(b) Write the alternative energy sources and its importance.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Write an account on mineral resources.

Or

(b) Elaborately discuss pond as an ecosystem.

17. (a) Give a detailed account on renewable and non-renewable energy sources.

Or

(b) Write an essay on types of biodiversity conservation.

18. (a) Describe the impact of water pollution and its sources.

Or

(b) Give a detailed account on soil pollution.

———————

CP–7844

Sub. Code
31

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Science

PART II – ENGLISH – COMMUNICATIVE SKILLS

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is meant by communication?
2. Define non-verbal communication.
3. Describe relative clauses with examples.
4. Change sentences into gerund :
   (a) These walls need to be painted.
   (b) Most children enjoy to play.
5. Insert articles in the blanks :
   At end of ———— street stands ———— old house built of ———— stone.
6. Define adjective with a example.
7. How many vowels and consonants are there in English?
8. When do we use rising intonation?
9. What is scanning?
10. Give two reasons for importance of reading.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Write a short note on types of communication.

Or

(b) Give a short note on ways of challenging the barriers of communication.

12. (a) Write correct question Tag for the following sentences :

(i) We are working tomorrow

(ii) They have been to Japan

(iii) She is Italian

(iv) They did not go out last Sunday

(v) He will be going tonight.

Or

(b) Change the sentences as directed.

(i) The boy caught a fish. (change into passive voice).

(ii) The entire sketch of highway was paved by the crew (change into active voice)

(iii) I will clean the house every Saturday. (Change into passive voice).

(iv) Who taught you to ski? (change into passive voice)

(v) No one responded to my sales add. (change into passive voice).

13. (a) Fill in the blanks with the correct word given in the bracket.

(i) We went to see a ———— last night. (play/theater).

(ii) I won't buy it at that ————. It is too hight. (cost/price).

(iii) The sound of the huge ———— filled the Church. (instrument/organ)

(iv) The bride wore a long cream-coloured gown, and the groom's ———— was dark green. (dress/suit).

(v) The ———— of fixing this old car is so high that I may just buy a new one instead. (value/cost).

Or

(b) Fill in the blanks with correct adverb.

(i) We ———— play handball.

(ii) ————, we could not see mount Everest.

(iii) Rani reads book ———— afternoon.

(iv) Peter sang the song ————in the bathroom.

(v) I ———— go to threaters.

14. (a) What are called consonants sounds? Explain it briefly.

Or

(b) What are called the stressed and unstressed syllables in English?

15. (a) Why reading in so important?

Or

(b) Explain : SQ3R.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Write an essay on kinds of verbal and non-verbal communication.

Or

(b) Write an essay on types of clauses.

17. (a) Use appropriate article to fill in the blanks.

(i) We are all ———— friends.

(ii) ———— honest man respects everywhere.

(iii) ———— University makes the students perfect.

(iv) After ———— long day, ————cup of tea tasted particularly good.

(v) Please give me ———— hammer.

(vi) My mother is ———— honest woman.

(vii) I heard ———— interesting story.

(viii) He did not have ———— time to rent an apartment.

(ix) We need ———— light in this room.

(x) She is ———— united states senator.

Or

(b) What are the rules there in using intonation? Explain with examples.

18. (a) What is reading? Differentiate land and silent reading.

Or

(b) Write an essay on characteristics of good hand writing.

———————

CP–7845

Sub. Code
32

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017.

Aeronautical Science

THERMO DYNAMICS

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** questions.

1. Define heof transfer.

2. Define work transfer.

3. Define law of conservation of mass.

4. Define Boyle's law.

5. Define Charle's law.

6. Mention various stages of dual cycle.

7. Define air conditioning.

8. Mention the formula to calculate the minimum air required for combustion.

9. Mention the components of piston engine.

10. Define steady flow energy equation.

Section B (5 × 5 = 25)

Answer **all** questions.

11. (a) Explain first law and second law of thermodynamics with research.

Or

(b) Explain steady flow energy equation.

12. (a) Write short notes on heat engine.

Or

(b) Write short notes on heat pump.

13. (a) Write short notes on calorific value of fuels.

Or

(b) Write short notes on combustion of fuels.

14. (a) Derive the formula for calculating the isothermal efficiency of reciprocating air compressor.

Or

(b) Write short notes on types of system and surroundings.

15. (a) Explain the advantages and disadvantages of multistage compression.

Or

(b) Explain the working principle of gas turbine engine.

Section C

$(3 \times 10 = 30)$

Answer **all** questions.

16. (a) Explain work and various types of work.

Or

(b) Explain various applications of steady flow energy equation.

17. (a) Derive the air standard efficiency of Brayton cycle.

Or

(b) Explain the classification of piston engine.

18. (a) Explain the working of axial flow compressors with a sketch.

Or

(b) Explain advantages and disadvantages of closed cycle gas turbine over open cycle gas turbine and determine the thermal efficiency of gas turbine.

CP–7846

Sub. Code
33

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Science

FLUID MECHANICS AND HYDRAULIC MACHINES

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define Specific Gravity.

2. What are the types of Mechanical Gauges?

3. Short notes on Buoyancy.

4. Define Flownet.

5. Write notes on water Hammer.

6. What are the losses happen in pipe flow?

7. Write any two classification of Turbine.

8. What is centrifugal pump?

9. Write short notes on Hydraulic Torque converter.

10. Write short notes on Gear-Pump.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Define Pressure. What are the various types of pressure?

Or

(b) Explain about Pascal's Law.

12. (a) Find the volume of the water displaced and position of centre of buoyancy for a wooden block of width 2.5 m and depth 1.5 m. When it floats horizontally in water. The density of wooden block is 650 kg/m^3 and its length 6.0 m.

Or

(b) Derive Velocity Potential function.

13. (a) Derive Euler's equation.

Or

(b) Three pipes of lengths 800 m, 500 m and 400 m and of diameter 500 mm, 400 mm and 300 mm respectively are connected in series. These pipes are to be replaced by a single pipe of length 1700 m. Find the diameter of the single pipe.

14. (a) Force exerted by a jet on a hinged plate.

Or

(b) Draw reciprocating pump. What are the various parts in reciprocating pump?

15. (a) Explain about Hydraulic press.

Or

(b) Explain about Hydraulic lift.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Explain the various properties of fluid.

Or

(b) Stability of submerged and floating body.

17. (a) Explain any two applications of Bernoulli's equation.

Or

(b) Hydroelectric power plant (with neat sketch).

18. (a) Write short notes with sketch :

(i) Hydraulic accumulator

(ii) Hydraulic intensifier.

Or

(b) The difference in water surface level in two tanks which are connected by three pipes in series of lengths 300 m, 170 m and 210 m and if diameters 300 mm, 200 mm and 400 mm respectively, is 12 m. Determine the rate of flow of water if co-efficient of friction are 0.005, 0.0052, 0.0048 respectively (Neglect Minor Losses).

CP–7848

Sub. Code
35

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Science

AIRCRAFT CONSTRUCTION

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Write the significance of fail-safe design?

2. Differentiate between wing rib and wing butt rib?

3. Define fly-by-wire system.

4. What is the function of Krueger flaps?

5. What is shock strut?

6. What is the function of Brake accumulator?

7. How will you calculate C.G in percentage of MAC?

8. What is the need for reweighting?

9. Mention the procedure involved in leveling of aircraft?

10. Write the sequence of rigging check and explain the aircraft engine alignment check?

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Describe the general construction of fuselage and its components.

Or

(b) Explain aircraft zoning and its purpose with examples.

12. (a) Write short notes on secondary flight control surfaces.

Or

(b) Explain the control around vertical axis.

13. (a) Explain the working of master cylinder in aircraft brake system with a neat sketch.

Or

(b) Explain Segmented Rotor Brake unit construction and its operation.

14. (a) Explain the theory of weight and balance.

Or

(b) Explain the working of Electronic weighting equipment.

15. (a) List out the guidelines involved in control surface rigging.

Or

(b) What are the necessary parameters that have to be considered while performing Biplane and fuselage alignment?

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Briefly explain the semi-monocoque fuselage construction in detail.

Or

(b) With the help of neat sketch, explain about the primary flight control surfaces.

17. (a) Explain in detail about the description of nose-wheel steering system?

Or

(b) Explain the procedures involved in Aircraft weighting?

18. (a) Explain the working of Anti-skid system with the help of schematic diagram?

Or

(b) Explain the procedures involved in rigging of flight control and performing symmetry checks.

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CP–7847

Sub. Code
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B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Aeronautical Science

AERODYNAMICS AND HELICOPTER THEORY

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define Bernoulli's principle.

2. Define pressure and humidity.

3. Explain parasite drag.

4. Define weight.

5. Define coridis effect.

6. Explain Spoilers.

7. Explain collective pitch.

8. Explain Havening.

9. Define Dampening.

10. What is translational lift?

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Explain Newton's laws of momentum.

Or

(b) Explain profile and terminology of aerofoil.

12. (a) Explain life-weight and drag-thrust couples.

Or

(b) Explain the fundamental equation for lift and drag.

13. (a) Explain blade flapping and conning.

Or

(b) Write short notes on gyroscopic precession and torque.

14. (a) Explain collective pitch, cyclic pitch and throttle control.

Or

(b) Write notes on stabilizer bar and swashplates.

15. (a) Write notes on tail rotor drive shaft.

Or

(b) Briefly explain tail gear box.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Explain the physical properties of air and their effects on aircrafts.

Or

(b) Explain longitudinal stability.

17. (a) Explain the fixed and movable control surfaces of an aircraft.

Or

(b) Explain the dissymmetry of lift.

18. (a) Write brief notes on blade alignment, static and dynamic balancing.

Or

(b) Explain pitch change mechanism and its operations.